

Autonomous Chemistry and Materials Innovation Driven by Scientific Agents

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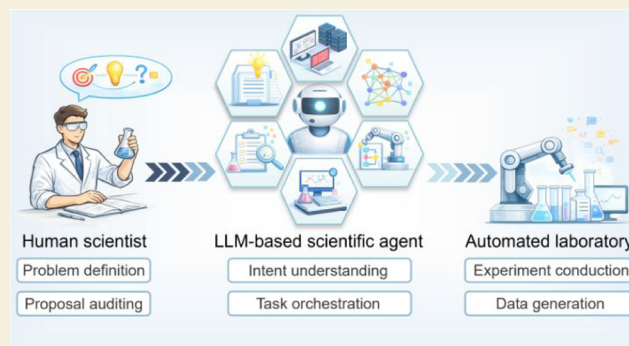
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ABSTRACT: Large language model (LLM)-based agents are reshaping how self-driving laboratories (SDLs) may support autonomous chemical and materials research. Although SDLs have enabled major advances in mechanized experimentation and closed-loop optimization, their scientific utility remains limited when tasks require literature-grounded reasoning, adaptive coordination, and interpretation beyond predefined search spaces. In this perspective, we examine how LLM-based agents may help bridge this gap by translating scientific intent into machine-executable workflows. We propose a five-module framework—Comprehension, Design, Execution, Analysis, and Optimization—to organize the capabilities required for agent-enabled SDLs, and we discuss representative systems, including Coscientist, ChemCrow, LLM-RDF, and AI-Chemist, as milestones in this transition. We also emphasize that agent-enabled SDLs should not be conflated with autonomous scientific discovery. Safety in physical execution, hardware interoperability, reproducibility, and auditability remain central challenges. To support a more critical assessment, we introduce the HYDRA framework for benchmarking trustworthy agent-enabled workflows. Finally, we outline a human–AI–SDL collaborative model in which scientists remain responsible for scientific framing, interpretation, and oversight.

KEYWORDS: autonomous chemistry, self-driving laboratories, scientific agents, large language models



1. INTRODUCTION

Autonomous chemistry can be viewed as an effort to shift parts of chemical research from manual, researcher-dependent operation toward machine-enabled workflows in which execution, data handling, and selected decision steps are carried out under explicitly defined human goals and constraints. In this framework, self-driving laboratories (SDLs), also referred to here as autonomous laboratories, have emerged as the key experimental infrastructure for mechanizing chemical tasks through integrated robotic synthesis, characterization, and data handling.¹ These systems have greatly improved throughput and operational consistency and have made closed-loop experimentation increasingly practical in chemistry and materials science. Yet, the mechanization of experimental tasks addresses only one layer of autonomy.² In their present form, SDLs are most effective for problems that can be formulated within relatively well-bounded experimental spaces, supported by modular workflows, and evaluated with measurable objective functions such as reaction optimization, catalyst screening, formulation design, and repetitive synthesis-characterization cycles. Many important scientific questions remain less amenable to SDL-based execution, particularly those that rely heavily on tacit

experimental knowledge, require material-specific protocol redesign, or demand mechanistic interpretation that cannot be reduced to a single optimization target.

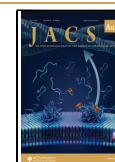
Within this operating regime, the cognitive core of most SDLs relies on traditional machine learning algorithms, such as Bayesian optimization (BO) or active learning strategies.^{3–5} While effective for optimizing specific parameters within rigid, predefined search spaces, these algorithms are limited in their ability to engage with a broader chemical context beyond structured numerical objectives. They are not designed to leverage literature to formulate initial hypotheses, troubleshoot hardware failures based on visual feedback, or dynamically replan experiments when faced with serendipitous or unexpected phenomena. Addressing these limitations in task planning, coordination, and adaptive decision-making will

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require a higher-level orchestration layer capable of linking scientific intent to executable laboratory actions.

Recent advances in large language models (LLMs), exemplified by systems such as ChatGPT,⁶ together with the emergence of scientific LLM-based agents⁷ (hereafter, agents), have made such a layer increasingly conceivable, creating new opportunities to broaden the functional scope of SDLs beyond execution-centered automation.⁸ We define an agent as an LLM-driven AI system capable of autonomously planning, executing, and analyzing scientific tasks by integrating reasoning, tool usage, and environmental interactions to manage end-to-end scientific workflows. Distinct from traditional discriminative models such as GNNs⁹ and SVMs¹⁰ that process structured data, agents are distinguished by their ability to reason with unstructured knowledge and orchestrate tools to both observe and manipulate the physical world.^{11,12} Viewed in this context, the emergence of agents can be understood as part of a broader evolution toward more autonomous research workflows. As illustrated in Figure 1, this

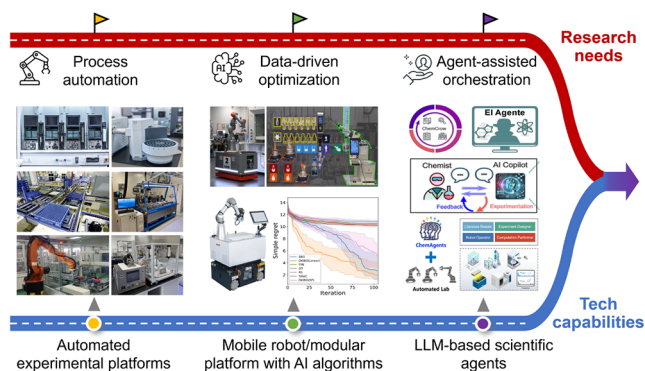


Figure 1. Conceptual coevolution of research needs and technological capabilities in chemistry and materials science. The landscape progresses from Stage I, high-throughput process automation based on robotic and instrumentation platforms, to Stage II, data-driven optimization enabled by modeling and closed-loop experimental cycles,^{13–16} and toward Stage III, agent-assisted orchestration supported by LLM-based systems integrated with SDLs.^{11,12,17,18} The upper trajectory indicates increasingly complex research demands, while the lower trajectory indicates expanding technological capabilities. Reproduced from ref 13, Copyright 2024, Springer Nature; ref 14, Copyright 2024, Springer Nature; ref 15, Copyright 2023, Springer Nature; ref 16, Copyright 2023, IOS Press; ref 11, Copyright 2024, Springer Nature; ref 12, Copyright 2025, American Chemical Society; ref 17, Copyright 2025, Elsevier; ref 18, Copyright 2025, American Chemical Society.

trajectory can be interpreted through the coevolution of research needs and technological capabilities in chemistry and materials science, giving rise to three broad stages:

1.1. Stage I: High-Throughput Process Automation

Represented by fixed-function automated platforms, this stage focused on accelerating repetitive experimental tasks through high-throughput screening systems, liquid handling platforms, and related robotic infrastructure.^{19,20} Although highly efficient, these systems remained largely rigid in workflow structure; deviations such as parameter adjustments, protocol modifications, or exploration of new reaction spaces still required substantial human intervention. As a result, Stage I platforms were particularly well suited to standardized and

repetitive operations but less adaptable to exploratory research settings.

1.2. Stage II: Data-Driven Optimization

With the integration of algorithms such as BO, SDLs acquired a limited but important capacity for iterative decision-making, enabling closed-loop design-make-test-analyze cycles.^{13,21} In this stage, robotics provided the physical execution layer, while machine learning supported numerical optimization within predefined experimental spaces. However, because these approaches operate primarily on structured objectives and parametrized search domains, their utility remained largely confined to bounded optimization tasks and was less suited to open-ended scientific problems requiring broader contextual reasoning or dynamic workflow adaptation.

1.3. Stage III: Agent-Assisted Orchestration

The recent emergence of LLM-based agents points toward a new stage in which SDLs may begin to support a broader range of research workflows. Rather than representing simply another optimization algorithm, agents introduce the possibility of a higher-level orchestration layer that can connect scientific goals to literature retrieval, task decomposition, tool coordination, and experimental execution. In this paradigm, researchers may specify higher-level objectives, such as identifying a highly soluble electrolyte, while the agent supports the translation of these objectives into executable workflows spanning information gathering, experimental planning, and iterative validation.

Accordingly, this perspective focuses on this emerging frontier. Our aim is not to frame agents as replacements for researchers but to examine how their integration with robotic laboratory systems may broaden the functional scope of SDLs beyond execution-centered automation. The following sections discuss the architectural principles required to incorporate agents into SDLs, analyze the challenges of grounding AI systems in physical laboratory environments, and outline the opportunities and limitations that may shape their future development in chemistry and materials science.

2. AGENT-BASED ARCHITECTURE IN SELF-DRIVING LABORATORIES

The transition from automated to autonomous laboratories relies on integrating a cognitive layer capable of managing the complexity. To understand how to construct this layer, we first examine the evolutionary trajectory of AI agents and how their capabilities map to the physical requirements of SDLs.

In artificial intelligence, the concept of an agent refers to an entity that perceives its environment and takes actions to achieve specific goals. Early agents, primarily in the field of Reinforcement Learning,⁴ excelled in closed-system games such as AlphaGo²² but struggled to cope with the open-ended ambiguity of the real world. The emergence of LLMs fundamentally altered this landscape. Initially functioning as static knowledge bases, LLMs evolved into reasoning engines with the introduction of frameworks such as ReAct²³ and Toolformer.²⁴ These innovations enabled LLMs to transcend the limitations of pure text generation, equipping them with the ability to decompose complex queries and invoke external digital tools such as search engines and calculators to solve problems. This evolution is particularly rapid in chemistry. Systems such as ChemCrow¹¹ and Coscientist²⁵ demonstrated that LLM agents could autonomously plan synthesis routes by querying databases like PubChem²⁶ and executing code scripts. However, while these software agents demonstrated impressive

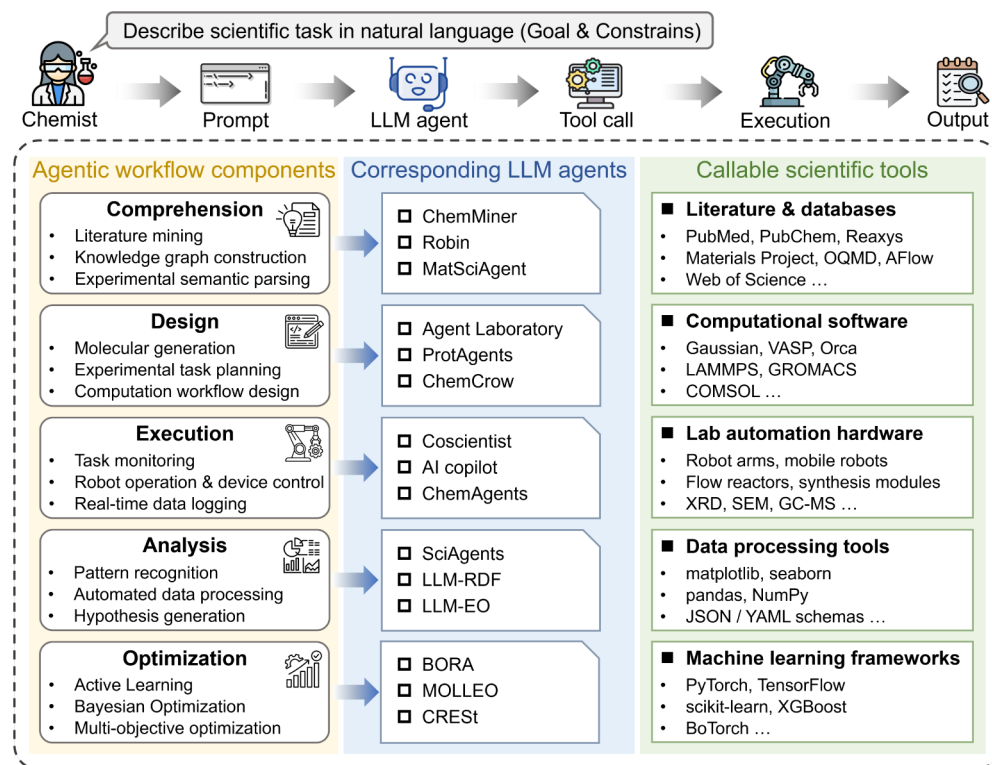


Figure 2. Agentic modules for self-driving laboratories. A scientific goal and its constraints are first specified in natural language and then translated into an agentic closed-loop workflow. The workflow is organized into five functional components: Comprehension, Design, Execution, Analysis, and Optimization, which, respectively, support literature/knowledge grounding, experiment and computation planning, robotic and instrument control with real-time logging, data processing and pattern recognition, and active-learning/Bayesian (including multi-objective) optimization. For each component, representative LLM-agent systems are listed (middle panel),^{11,17,18,25,27–37} highlighting how specialized agents can be composed into a coordinated pipeline. The right panel summarizes typical callable scientific tools that agents interface with, spanning literature databases, computational software, lab automation hardware, data-processing utilities, and machine-learning frameworks.^{38–48} Together, these modules provide the “brain–hands–eyes” of an embodied SDL by coupling reasoning, tool orchestration, and physical experimentation in a traceable closed loop.

reasoning capabilities, they remain largely disembodied, capable of planning experiments but still reliant on humans for physical execution. To realize true autonomous chemistry, we must bridge the gap between these digital reasoning agents and the physical hardware of SDLs.

Constructing an agent-driven SDL requires more than simply connecting a chatbot to a robotic arm. It demands a systematic architecture designed for closed-loop research centered on a supervisory orchestration and coordination hub that organizes dispersed scientific tools and experimental resources into functional modules (Figure 2). These modules include Comprehension, Design, Execution, Analysis, and Optimization. In the context of an embodied SDL, these components collectively constitute the brain, hands, and eyes of the system.

Comprehension, Design, and Optimization jointly undertake goal parsing, knowledge integration, scheme generation, and iterative decision-making, corresponding to the brain of the system. Before experimentation begins, the agent must ground its actions in chemical reality and verifiable evidence. The Comprehension module, exemplified by ChemMiner²⁷ and LLM-RDF,³³ typically employs Retrieval-Augmented Generation (RAG)⁴⁹ to extract key information from unstructured literature, construct structured knowledge graphs, and parse research intents and constraints. For an SDL, this implies that the system does not proceed through trial and error from scratch but proposes credible hypotheses and feasible search

directions based on accumulated chemical knowledge. Building on this, the Design module acts as a generation and planning engine. Transcending fixed templates, it translates hypotheses into executable experimental schemes. For instance, Agent Laboratory³⁰ and ProtAgents³¹ demonstrate how agents organize computational workflows, generate molecular or material representations (SMILES/CIF), and select synthesis routes. Simultaneously, the planner explicitly incorporates physical and resource constraints such as reagent availability and equipment capability windows, thereby bridging the gap between theoretical feasibility and experimental practicability. To close the loop, the Optimization module updates strategies and internal belief models based on analytical feedback. Agent-enhanced methods represented by BORA³⁵ integrate semantic reasoning and reflection into BO. This enables the system not only to optimize numerical metrics but also to provide interpretable rationales for results, adjust search spaces, and trigger subsequent design decisions, elevating iteration from parameter search to knowledge-driven exploration.

Execution is responsible for compiling plans into executable protocols and driving devices to complete experimental operations, corresponding to the hands of the system. It serves as the critical interface transforming digital decisions into physical actions and represents the core intersection of the agent and the embodied SDL. The agent drives automation hardware, such as robotic arms, mobile robots, and flow reactors, through functional APIs. The difficulty lies in

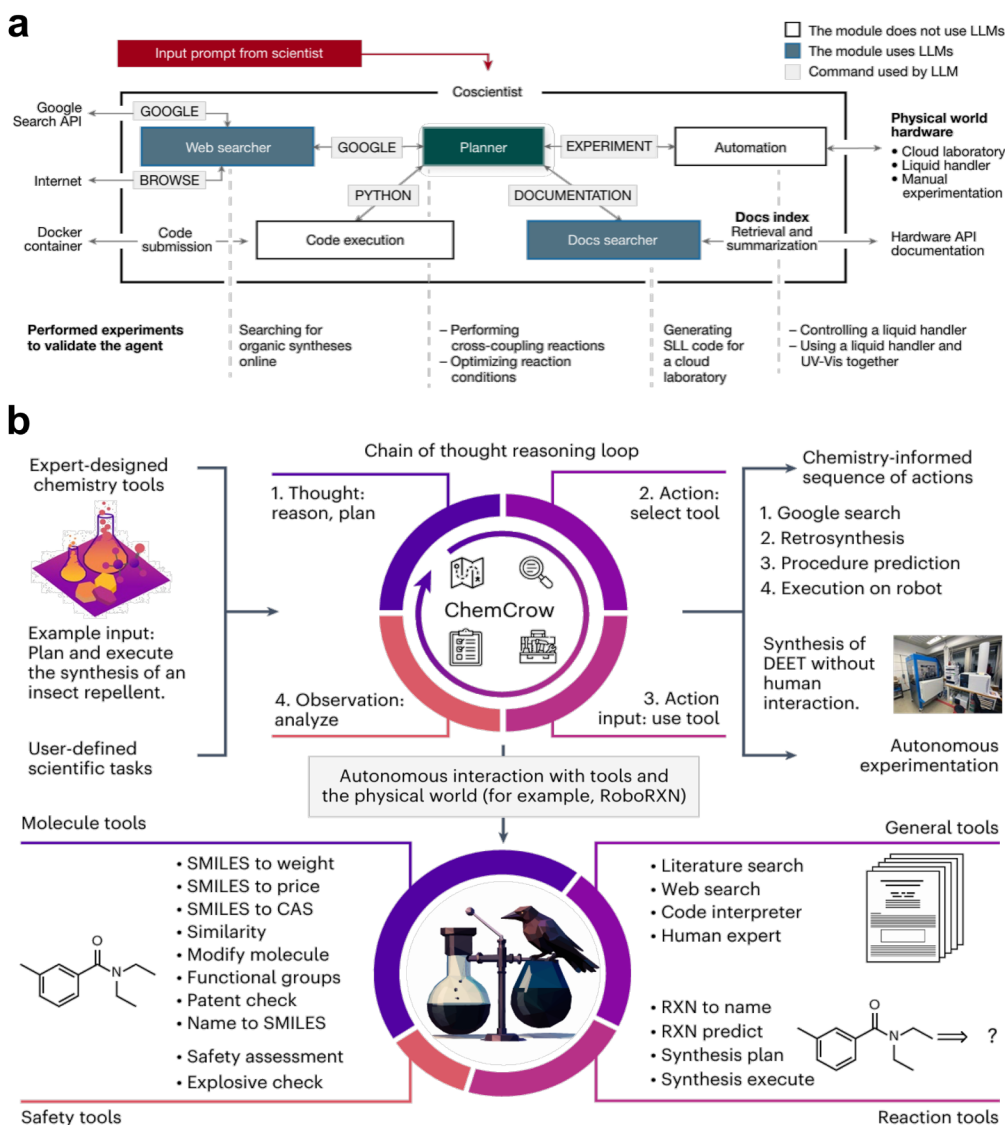


Figure 3. a. System's architecture of Coscientist.²⁵ Reproduced from ref 25, Copyright 2023, Springer Nature. b. Overview and expert-designed toolset of ChemCrow.¹¹ Reproduced from ref 11, Copyright 2024, Springer Nature.

achieving a stable semantic-to-physical mapping, which translates high-level intents like heating to reflux or adding dropwise to a specific pH into verifiable and replayable sequences of device instructions, such as temperature control, stirring, liquid handling, and wait conditions. It must also handle scheduling, interlocking, and exceptions during execution. Systems such as Coscientist²⁵ and ChemAgents¹² demonstrate feasible pathways for autonomously executing complex physical tasks by enabling agents to read hardware documentation, invoke hierarchical subagents, or adopt structured action representations.

An autonomous system must perceive the consequences of its actions. Analysis extracts metrics from instrument and sensor data, performs quality control and uncertainty assessment, and feeds structured feedback into the closed loop, corresponding to the eyes of the system. This module combines data processing tools like pandas and numpy with multimodal perception.^{44,46} Beyond mere recording and plotting, it emphasizes the online and offline understanding of the experimental state. This includes visual cognition via cameras to detect color changes, precipitation, or interface

states alongside algorithms for anomaly detection, quality control, and uncertainty evaluation. This process converts raw signals into structured metrics and evidence that can flow back into the brain. As embodied by frameworks like Robin,²⁸ this module enables the SDL to form higher-level explanations and inferences from observations, providing reliable feedback for subsequent optimization rather than merely accumulating data.

By organic integration of these five modules, the SDL transforms from a rigid automated machine into an adaptable and intelligent partner. It no longer functions merely as a tool executing human-prescribed instructions but evolves into a full-process scientific discovery engine capable of independently navigating the literature from reading to hypothesis validation. This marks a decisive step for chemical research toward true zero-intervention autonomy.

3. EVOLUTION OF END-TO-END CHEMICAL RESEARCH AGENTS

The five-module architecture depicted in Section 2 is not a mere conceptual abstraction but a structured synthesis of key recent advances in scientific agents. From an observational

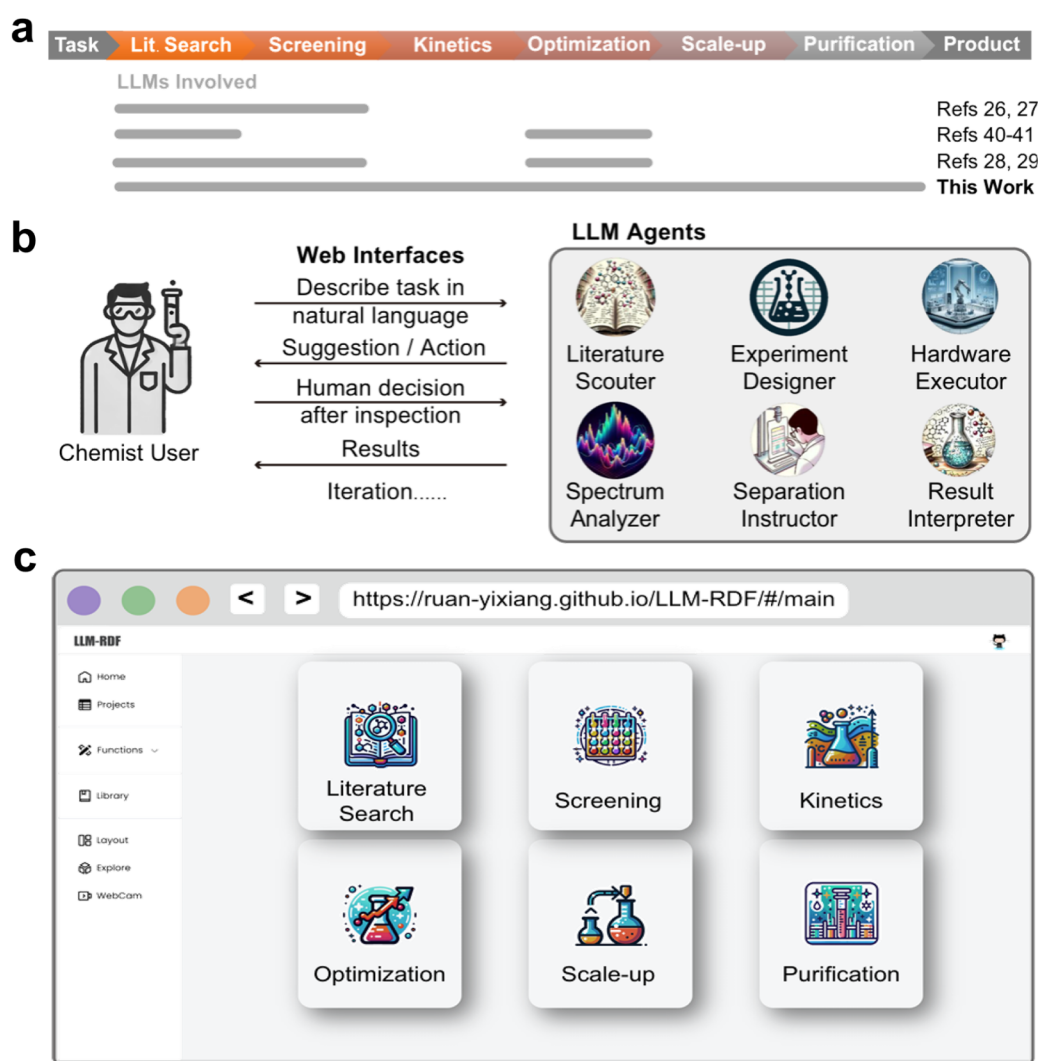


Figure 4. Overview of LLM-based multi-agent system for reaction development.³³ **a.** Workflow for chemical synthesis development facilitated by LLMs. **b.** Diagram illustrating the interactions between human chemists and the agents for performing tasks in chemical synthesis development. **c.** The web application with agents as the backend for end-to-end reaction development. Reproduced from ref³³, Copyright 2024, Springer Nature.

perspective, the development of these systems can be summarized as initially filling critical gaps in cognition, tooling, and execution at individual points before progressively moving toward end-to-end closed loops with cross-stage coordination. We focus on three representative works: Coscientist, ChemCrow, and LLM-RDF. These works do not merely represent alternative approaches; they signify a progressive lighting up of the architectural map. Coscientist first bridged the gap between Design and Execution, validating the feasibility of agent-driven hardware.²⁵ ChemCrow subsequently strengthened Comprehension and Design, introducing tool-augmented rigor to mitigate hallucinations.¹¹ Finally, LLM-RDF integrated a distinct Analysis module via multiagent collaboration, closing the loop from experimental data back to hypothesis generation.³³ By analyzing this progression, we illustrate how the scattered capabilities of agents coalesce into a unified system capable of genuine scientific discovery.

The primary barrier to autonomous chemistry was enabling a text-based model to interact with the physical world, a challenge centered on the Execution module. As shown in Figure 3a, Coscientist served as the critical proof of concept for this capability. By reading technical manuals and APIs for

liquid-handling robots, the Coscientist successfully translated high-level chemical goals, such as palladium-catalyzed cross-couplings, into precise machine-executable code. Its contribution was establishing that an agent could function as an adaptive planner capable of in-context learning to control unfamiliar hardware while implementing a tool calling for basic Optimization. However, regarding the Comprehension module, Coscientist relied primarily on the intrinsic knowledge of the foundation model (GPT-4). Without specialized verification tools, it remained susceptible to hallucinations, and its capability to perform deep Analysis of complex spectral data was limited, restricting its potential for mechanistic insight.

To address the reliability of the brain, ChemCrow focused on fortifying the Comprehension and Design modules through a tool-use paradigm (Figure 3b). Recognizing that pure LLMs struggle with precise chemical facts, ChemCrow integrated 18 expert tools, ranging from literature mining (LitSearch) to safety assessment (GHS checks) and molecular property calculation (RDKit). Instead of relying on memory, the agent acts as a manager that queries validated databases before generating synthesis plans, ensuring that its designs are chemically plausible and safe. While ChemCrow successfully

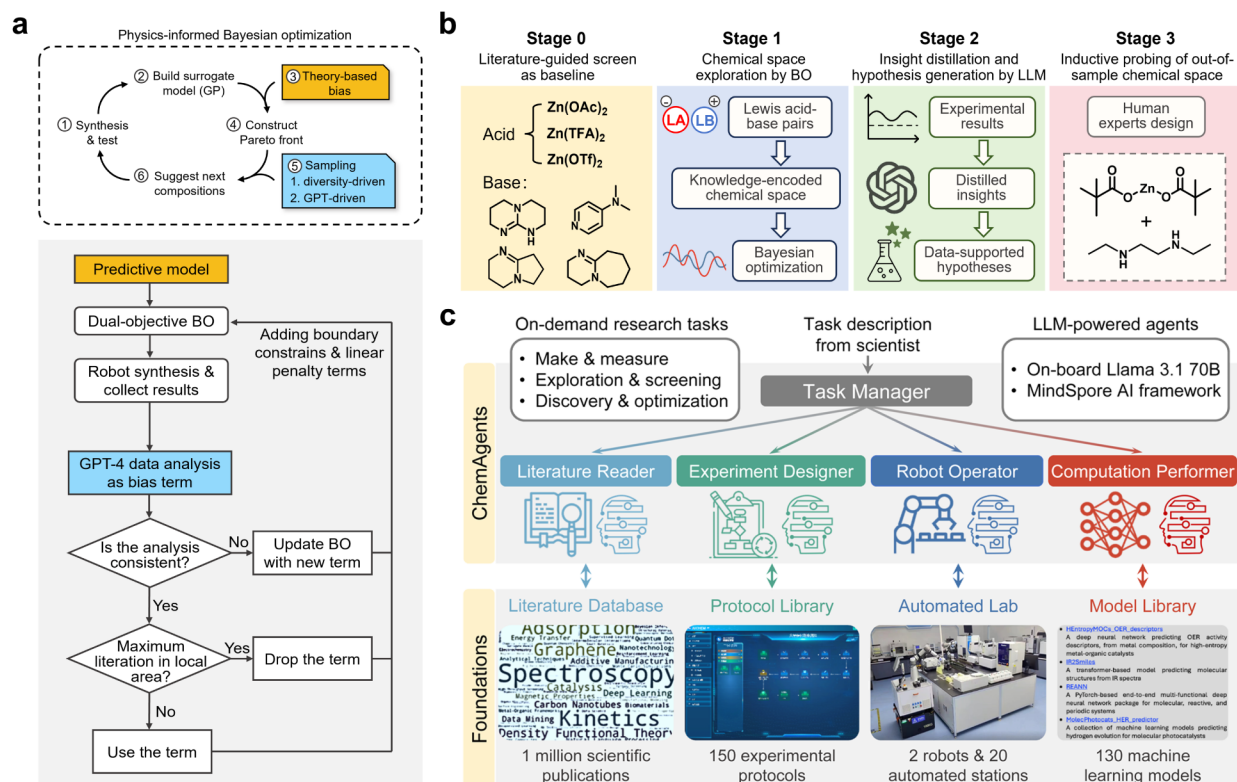


Figure 5. a. Physics-informed Bayesian optimization workflow combining theoretical models and LLM-based pattern analysis.⁵¹ Reproduced from ref 51, Copyright 2025, Elsevier. b. AI-guided framework for accelerated discovery of Lewis acid–base catalysts for PET glycolysis.⁵² Reproduced from ref 52, Copyright 2026, American Chemical Society. c. The LLM-based multiagent system, ChemAgents.¹² Reproduced from ref 12, Copyright 2025, American Chemical Society.

connected to a robotic platform for execution, its workflow remained largely linear and task-specific. It excelled at generating reliable recipes but lacked a robust Analysis module to autonomously interpret experimental feedback and trigger iterative Optimization cycles, leaving the research loop open.

The latest evolution, LLM-RDF, aims to close this loop by deploying a multiagent architecture that explicitly incorporates Analysis and Optimization (Figure 4 a–c). Unlike previous systems that focused on planning or synthesis, LLM-RDF introduces specialized agents for result interpretation, capable of processing raw instrument data, such as GC-MS and NMR spectra. This visual and numerical perception allows the system to assess the reaction yield and purity autonomously. In demonstrations of catalytic reaction optimization, the system achieved a closed loop, where the Analysis module fed insights back into Design, allowing the agent to iteratively adjust experimental parameters without human intervention. While LLM-RDF represents a structural prototype of a complete SDL, its current implementation relies on preprompted roles and defined reaction templates. The future challenge lies in generalizing this architecture to support open-ended exploration, where the agent defines its own research questions.

4. CASE STUDY: EMPOWERING AI-CHEMIST WITH SCIENTIFIC AGENTS

To concretize the implementation pathway of the agent architecture discussed above, this section presents a case study of the AI-Chemist platform developed by the University of Science and Technology of China shown in Figure 5. Prior to the integration of large language models, this platform

functioned as a standardized hardware ecosystem driven by numerical optimization algorithms.^{15,50} Building upon this robust physical infrastructure, this section focuses specifically on the technological leap toward agentification. We illustrate how the introduction of a cognitive layer transforms the platform from a high-throughput automated machine into an intelligent partner capable of physics-informed reasoning and on-demand task orchestration.

4.1. LLM-Enhanced Exploration

Initially, the integration of LLMs into the AI-Chemist platform functioned primarily as a source of supplementary information rather than as agents capable of directly manipulating hardware or computational tools. The first implementation addressed the challenge of multiobjective optimization in high-entropy alloy nanzyme design by introducing a physics-informed dual-objective BO framework (Figure 5a).⁵¹ Crucially, this stage implemented an LLM-in-the-loop mechanism, where models like GPT-4 served as online analysis components. By providing interpretable feedback based on data distribution and correlation patterns, the LLM dynamically influenced sampling preferences, moving optimization from a purely numerical drive toward a controllable exploration synergistically driven by knowledge and data. Subsequently, LLMs were applied to hypothesis generation in the Lewis acid–base catalyzed glycolysis of PET (Figure 5b).⁵² We designed a three-stage intelligent optimization workflow: BO first expanded the search space; the LLM then proposed verifiable hypotheses based on historical data; and finally, the system identified optimal catalyst formulations outside the original search space based on these validated hypotheses. These works demon-

strated the potential of cognitive augmentation in analysis and reasoning, laying the essential foundation for subsequent full agentification of the platform.

4.2. Agentic Orchestration

The second phase marked the transition from cognitive augmentation to full agentic orchestration. The proposal of the ChemAgents multiagent system embodied this shift, upgrading the platform from specialized workflows to on-demand task orchestration for open-ended instructions (Figure 5c).¹² Distinct from previous implementations where LLMs served as advisors, ChemAgents established a hierarchical multiagent architecture using onboard open-source large models (e.g., Llama-3) as the core reasoning engine.⁵³ A Task Manager coordinates role-specific agents—including a Literature Reader, Experiment Designer, Computation Performer, and Robot Operator—to decompose natural language requirements and orchestrate multistep execution. This decoupled architecture allows the system to handle open-ended instructions and reorganize workflows flexibly across different hardware modules, realizing an embodied SDL capable of delivering organizable, scalable, and responsive research on demand.

In summary, the integration of LLMs and multiagent architectures fundamentally redefines the capabilities of the AI-Chemist platform. Transcending the limitations of optimization within fixed search spaces, the agent-driven phase introduces a semantic and organizational layer that liberates the SDL from rigid workflows. LLMs and agents serve the platform by bridging the gap between abstract scientific intent and concrete hardware execution, enabling the system to comprehend natural language goals, autonomously decompose complex tasks, and orchestrate heterogeneous tools without hard-coded scripts. This shift resolves the long-standing bottleneck of operational rigidity in automation, transforming the SDL from a high-throughput tool into a collaborative partner capable of interpretable reasoning, flexible task reconfiguration, and on-demand autonomous research.

5. FUTURE PROSPECTS AND CHALLENGES

5.1. Limitations and Systemic Bottlenecks

The integration of LLM-based agents into scientific research offers new possibilities for knowledge aggregation, workflow automation, and traceable decision support. However, as these systems move from text processing to interaction with physical laboratory environments, important limitations in their underlying capabilities become more consequential. In this setting, the central risks are no longer limited to textual hallucinations but extend to failures that propagate into real-world experimental operations. These challenges are particularly evident in three dimensions.

The first and most immediate concern is algorithmic hallucination or semantic misinterpretation, which could translate into unsafe physical actions. Although state-of-the-art LLMs perform strongly in language generation, their lack of physical grounding makes them vulnerable to errors that are unacceptable in a laboratory setting. When an agent is permitted to influence or control physical execution, even small interpretive mistakes may lead to serious consequences, such as the mixing of incompatible reagents or the initiation of runaway exothermic reactions. For example, a literature instruction such as “add reagent B slowly due to a highly exothermic reaction” contains substantial tacit experimental knowledge. An LLM may oversimplify “slowly” into an

arbitrary fixed flow rate, such as 1 mL min⁻¹, which could still be dangerously fast for a sensitive or scaled-up system. Accordingly, future SDL architectures should incorporate deterministic safety constraints within the execution layer as a nonbypassable final safeguard. These constraints should retain the authority to override agent-generated commands, ensuring that probabilistic reasoning never exceeds the physical safety limits.

A second major bottleneck lies in the fragmented and largely closed hardware ecosystems of modern laboratories. To coordinate physical instruments autonomously, scientific agents require not only software access through APIs but also sufficiently detailed and machine-readable operational documentation. In practice, however, many advanced scientific instruments remain proprietary and poorly interoperable. As a result, researchers are often forced to build bespoke adapters for each device, which limits portability across laboratory environments and constrains the generalizability of agent-based workflows. Addressing this bottleneck will require closer coordination between the chemistry community and instrument vendors to establish more standardized communication protocols, interoperable control layers, and unified data formats. Such interoperability is not merely an engineering convenience; it is a prerequisite for a broader deployment of agent-enabled SDLs.

A third challenge concerns scientific reproducibility and data provenance. LLM-orchestrated workflows introduce a distinctive tension: while the physical sequence of actions may be replicated, the reasoning that produced those actions may not be fully stable across sessions. Because LLMs are probabilistic systems, identical inputs and goals do not always guarantee identical reasoning paths or intermediate decisions. This variability complicates transparency and weakens reproducibility at the level of digital decision-making. For this reason, future SDL architectures should incorporate rigorous audit trails that record prompts, model calls, tool invocations, intermediate outputs, and raw instrument traces. Such records can provide an auditable account not only of what the laboratory did but also of how the system arrived at those actions.

5.2. Benchmarking Agent-Assisted SDLs

To distinguish genuinely open-ended scientific investigation from automated hyperparameter tuning and to avoid inflated claims in a rapidly evolving field, the community needs more appropriate evaluation frameworks. Historically, the performance of Stage I and Stage II SDLs has often been judged primarily by engineering metrics such as throughput, optimization speed, or final product yield. These measures remain important, but they are insufficient for evaluating systems that aspire to support broader research workflows. A workflow that screens 1,000 catalysts through a hard-coded grid search may produce a high-performing outcome, yet still contribute little to hypothesis testing, mechanistic understanding, or adaptive scientific reasoning.

To evaluate the scientific validity, safety, and operational robustness of agent-enabled SDLs more systematically, we propose the HYDRA framework, comprising five dimensions: Hypothesis-testing readiness, Yielding and safety interlocks, Data governance, Reproducibility, and Auditability.

First, **Hypothesis-testing readiness** requires that agentic workflows move beyond linear search and support rigorous experimental design, including appropriate controls, random-

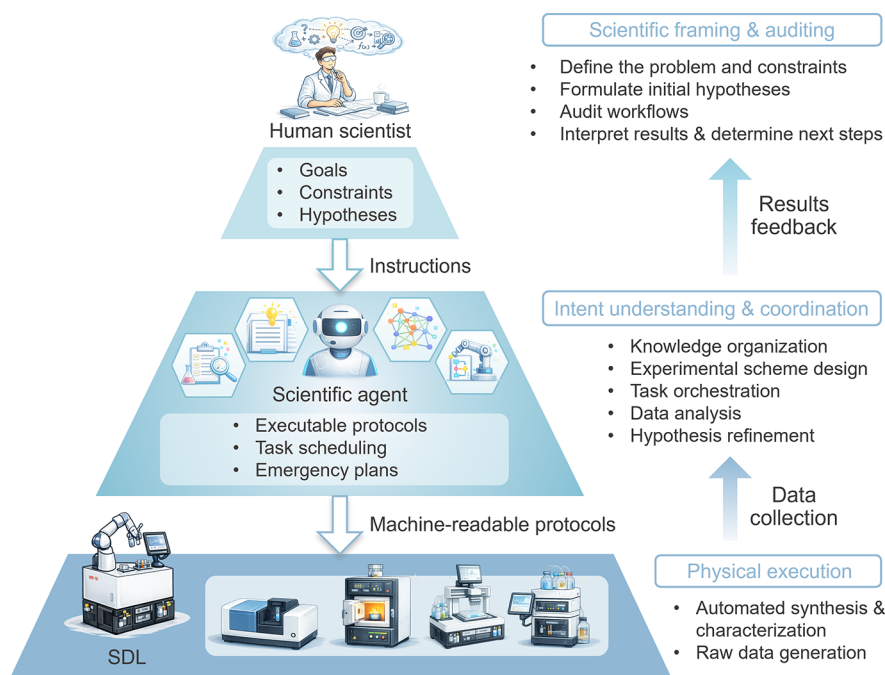


Figure 6. Hierarchical orchestration of human scientists, scientific agents, and SDLs in autonomous chemistry. Human researchers define goals, constraints, and hypotheses and audit workflows and results. Scientific agents translate these objectives into executable protocols through knowledge organization, experimental design, task orchestration, and data analysis. SDLs execute machine-readable protocols, perform automated synthesis and characterization, and generate raw experimental data. Data and results then flow back upward to support interpretation, auditing, and iterative decision-making.

ization, replication, and cross-validation where relevant. Second, **Yielding and safety interlocks** refer to the mechanisms by which an autonomous system must defer to deterministic safeguards or human oversight when anomalous data, hazardous conditions, or uncertain actions are encountered. Third, **Data governance** requires a transparent and traceable pathway from raw instrument signals to decision-grade data products with explicit quality-control rules, anomaly flagging, and uncertainty assessment. Fourth, **Reproducibility** extends beyond code release and requires detailed recording of protocol versions, reagent batches, calibration status, and environmental conditions. Finally, **Auditability** requires an action-level chain of custody across the workflow, including which agent or module made a decision, what evidence was used, what tools were called, how the action was tracked, and where human review entered the loop.

Taken together, the HYDRA framework is intended to provide a practical baseline for evaluating whether agent-enabled SDLs function as trustworthy research systems rather than opaque trial-and-error engines.

5.3. Toward Human–AI Collaboration in Autonomous Chemistry

The integration of agents into physical laboratories is not a single technical event but a gradual process involving increasing autonomy in decision-making, broader tool orchestration, and more tightly auditable closed-loop workflows. In this sense, the incorporation of agents into SDLs is not merely a software upgrade; it also reshapes how scientific labor is distributed across human researchers, AI systems, and the laboratory infrastructure. Its success will depend on sustained dialog between domain scientists and AI researchers as well as on the evolving skill set required for effective scientific practice. This redistribution of responsibilities can be

understood as a layered division of labor across the autonomous chemistry workflow (Figure 6). Human scientists remain responsible for high-level scientific framing and auditing; agents function as an intermediate layer for intent understanding, knowledge organization, and task coordination; and SDLs provide the physical execution layer for automated experimentation and raw data generation.

One likely benefit of this transition is the reduction of low-level operational friction. By offloading repetitive tasks such as manual pipetting, hardware interfacing, and script maintenance to automated systems, researchers may reclaim cognitive bandwidth for higher-level scientific work, including defining problem boundaries, formulating hypotheses, designing controls, and interpreting unexpected physical behavior. Yet, this redistribution of labor may also carry a hidden cost: the degradation of foundational physical intuition. The current generation of researchers often possesses an important advantage because their expertise was forged through direct bench experimentation and manual workflow design. Such hands-on experience provides the practical physical intuition required to rigorously audit agentic reasoning and detect subtle but consequential hallucinations that may appear linguistically plausible yet are physically or chemically unsound. If future generations of chemists become overly dependent on high-level AI orchestration without meaningful exposure to tactile, trial-and-error experiments, they may become less capable of critically evaluating the physical plausibility of AI-generated workflows. Autonomous chemistry may therefore give rise to a scientific Matthew effect, whereby AI disproportionately empowers established experts with deep domain knowledge while potentially deskilling or misleading less experienced researchers who lack the background needed to interrogate algorithmic outputs. This possibility highlights the enduring role of the scientist as an auditor in the AI era and suggests that

physical laboratory training should remain a central anchor of scientific rigor rather than a dispensable precursor to automation.

Conversely, the continuous operation of SDLs may provide an important route for improving the agents themselves. Although foundation models are powerful language systems, they typically lack grounded representations of physical causality. SDLs can help address this limitation by generating rich multimodal records of experimental action, including robotic logs, sensor outputs, system states, and recovery trajectories.⁵⁴ Such data can support the development of more domain-specific world models and provide training signals for methods such as behavioral cloning or offline reinforcement learning.⁵ In this way, model training may gradually shift from predicting linguistically plausible continuations to better anticipating the physical consequences of laboratory actions. Because mainstream LLMs are trained primarily on static text corpora,⁵⁵ they often remain more reliable in linguistic synthesis than in numerical reasoning or physically grounded inference.^{56–58} Laboratory-derived trajectory data may therefore play a key role in improving long-horizon planning and action grounding in future agents.

Overall, the combination of agents and SDLs is pushing chemical research toward more integrated human–AI–robot workflows. Their long-term value lies not in replacing chemists but in enabling a collaborative architecture in which humans define scientific questions, hypotheses, and standards of evidence, while agents and laboratory systems contribute scalable coordination, execution, and data handling. In this view, the promise of autonomous chemistry lies in strengthening the connection between systematic experimentation and scientific reasoning rather than in removing researchers from the discovery process.

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Notes

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